



Department of Computer Science and Business Systems

Academic Year 2024 – 2025 (Even Semester)

Degree, Semester & Branch: III Semester B.Tech. CSBE

Course Code & Title: CCS372 & Virtualization

Name of the Faculty member (s): Mrs.M.Jeya Sundari, AP/CSBE

Innovative Practice Description

- **Unit / Topic:** Unit II / Server Virtualization
- **Course Outcome:** CO2
- **Topic Learning Outcome:** TLO2
- **Activity Chosen:** Concept Map
- **Justification:**

Concept map becomes a powerful tool for both teaching and revision, catering to visual learners and promoting a deeper grasp of server virtualization technology. Creating a concept map on *Server Virtualization* used an innovative and effective way to visually represent and understand the complex relationships between its key components such as hypervisors, virtual machines (VMs), resource allocation, and different virtualization types (like full, para, and hardware-assisted virtualization). It simplifies the understanding of technical processes like workload isolation, hardware abstraction, and cost-efficient infrastructure management.

Time Allotted for the Activity: 40 Minutes

- **Details of the Implementation:**
 - The activity was conducted at the last 10 minutes of the class after discussing the various Virtualization concepts.
 - The students are asked to prepare a concept map by depicting the different concepts involved in Virtualization.
 - The students recollect the concepts involved in Server Virtualization and present the concepts in a easily understandable way.
 - Fig.1 shows the concept maps drawn by the students.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	2	1	3	2	2	1	2	1	2	3	1	1

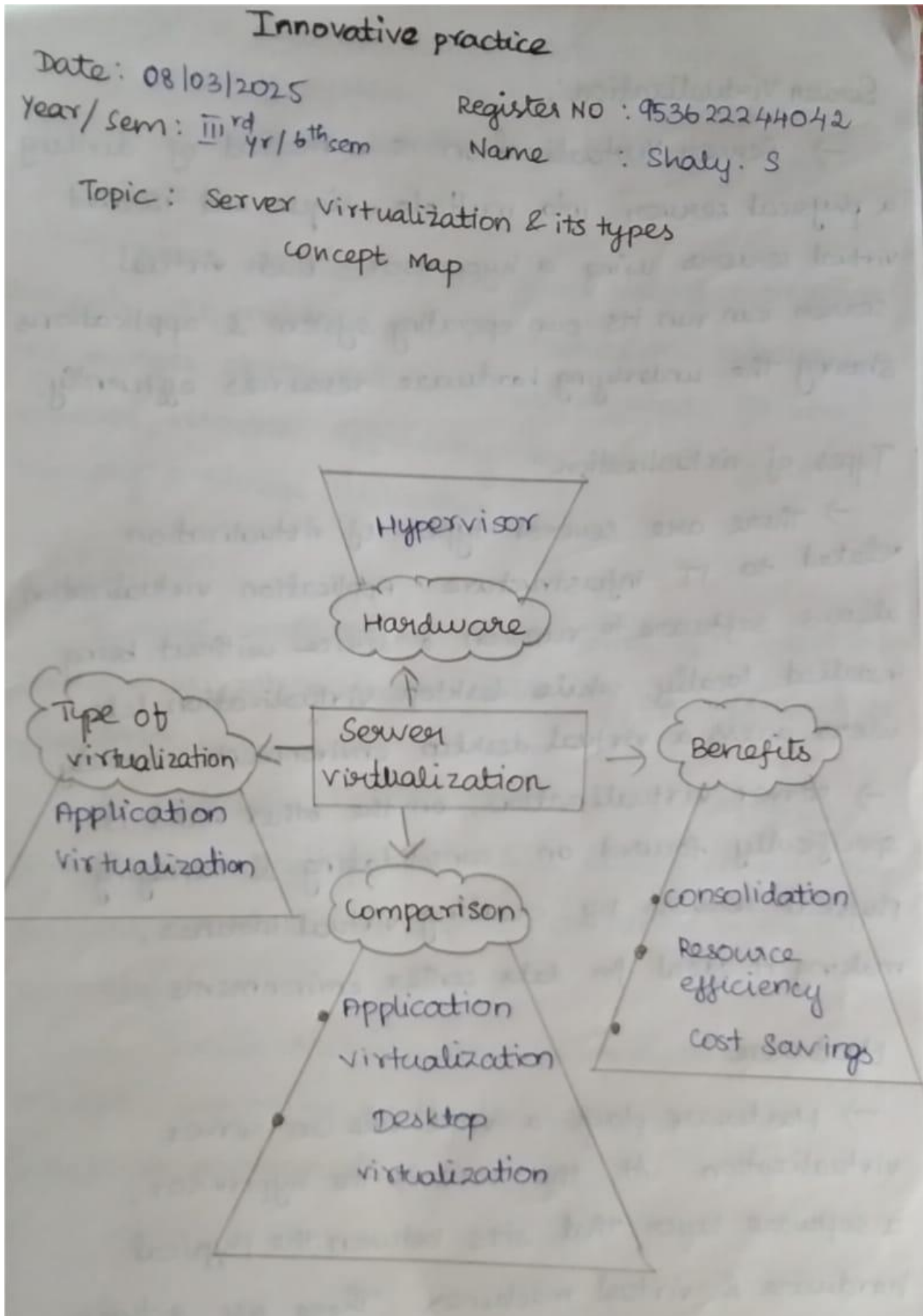
(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO5	PO6	PO8
	3	3	1	1	1	1	3
Justification for correlation	The students apply the knowledge of Networking and Operating System	The students identify the ways to represent the concepts of Virtualization	Students design virtual systems with proper resource allocation, fault tolerance, and n/w security	Students investigate system behavior by VM configurations, analyzing performance, and interpreting results for optimized solutions.	Students use modern tools to create and manage virtual environments.	Students assess societal and safety impacts of virtual systems in real-world contexts.	Students follow ethical practices while managing data and privacy in virtual systems.

Innovative practice	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
	2	1	1	3	3	1	1
Justification for correlation	Students collaborate in teams to design and troubleshoot virtual environment	Students clearly present and document virtual system designs and findings.	Students apply project management principles to plan, execute, and optimize virtualization projects.	Students develop skills for continuous learning and staying updated with emerging virtualization technologies.	Students use cutting-edge technologies to develop advanced virtual systems, ensuring efficient problem-solving and management.	Students use virtual systems and software models to solve real business problems.	Students apply management skills and real-time project experience to develop ethical virtual systems in computer science and business.

• Images / Screenshot of the practice:



Server Virtualization:

→ Server virtualization is a method of dividing a physical server into multiple unique and isolated virtual servers using a hypervisor. Each virtual server can run its own operating system & applications sharing the underlying hardware resources efficiently.

Types of virtualization:

→ There are several types of virtualization related to IT infrastructure. Application virtualization allows software to run on a device without being installed locally, while desktop virtualization lets users access a virtual desktop environment remotely.

→ server virtualization, on the other hand is specifically focused on consolidating & managing physical servers by creating virtual instances, making it ideal for data center environments.

Hardware:

→ Hardware plays a vital role in server virtualization. At the core is the hypervisor, a software layer that sits between the physical hardware & virtual machines. There are 2 types of hypervisors.

→ Type 1 (Bare-metal)

→ Type - 2 (hosted) which runs on top of an existing operating system. The hypervisor allocates resources & ensures isolation b/w virtual machines.

Benefits :

→ Server Virtualization offers several significant benefits. It enables server consolidation, reducing the number of physical machines required. It also improves resource efficiency by maximizing CPU, memory & Storage Utilization.

Comparison :

→ When compared to other forms of virtualization, server virtualization focuses primarily on infrastructure optimization.

→ Application virtualization isolates software from the operating system while desktop virtualization delivers complete desktop environments to users remotely.

→ Server Virtualization is more backend-focused, supporting workloads & services within enterprise environments.

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- All the students are very much interested in drawing the features of Server Virtualization as a concept map.

- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- This activity helps them to summarize the entire concepts in one picture.
 - The level of understanding can be analyzed by seeing the concept map they have drawn.

- ❖ ***Challenges faced in implementation:***

- Some of the students need more time to complete the map.
 - Some students only represent few concepts in the map.

References:

- <https://www.youtube.com/watch?v=ilILdQuer0E>
- https://www.ritrjpm.ac.in/images/computer-science/2021-2022/MGM_VI_concept_summarization.pdf

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